

Part A. PERSONAL INFORMATION

Part A. PERSONAL INFORMATION		CV date		09.07.2026
First name	Amirreza			
Family name	Mottafegh			
Gender	Male	Birth date	18.06.1994	
Passport number	F97805064			
E-mail	Armtfgh@postech.ac.kr	Armtfgh.github.io		
Open Researcher and Contributor ID (ORCID)		0009-0004-7090-845X		

A.1. Current position

Position	Postdoctoral Researcher		
Initial date	September 2025		
Institution	Korea Research Institute of Chemical Technology		
Department/Center	Chemical Platforms	Digital Chemistry Research Center	
Country	South Korea	Teleph. number	+821042201373
Key words	Automation, Digital Chemistry, Machine Learning, Self-Driving Labs, Microfluidics		

A.2. Previous professional status

Period	Position/Institution/Country
2021.08-2025.08	Graduate Student/Pohang University of Science and Technology/South Korea
2025.09-Present	Postdoctoral Researcher/Korea Research Institute of Chemical Technology/South Korea

A.3. Education

PhD, Licensed, Graduate	University/Country	Year
M.Sc./Ph.D. Integrated (Chemical Engineering Department)	Pohang University of Science and Technology, South Korea	2021-2025
M.Sc. (Energy Engineering Department)	Sharif University of Science and Technology, Iran	2018-2020
B.Sc. (Chemical Engineering Department)	Sharif University of Science and Technology, Iran	2012-2017

Part B. CV SUMMARY

Dr. Mottafegh currently serves as a Postdoctoral Research Fellow at the Korea Research Institute of Chemical Technology (KRICT), a highly selective position with only 1-2 annual appointments in the Digital Chemistry Center. In this role, he continues advancing self-driving laboratory platforms and AI-guided chemical synthesis. He completed his PhD at POSTECH (South Korea, 2021-2025) through the prestigious Brain Korea 21 (BK21) Fellowship. His doctoral research focused on developing Self-Driving Laboratories for nanomedicine, creating an Autonomous Integrated Microfluidic (AIM) platform that combined combinatorial synthesis, inline purification, and real-time characterization. By implementing Multi-Objective Bayesian Optimization algorithms, he accelerated nanoparticle optimization from days to ~20 minutes per experiment, executing ~1,000 autonomous synthesis loops with >80% encapsulation efficiency. This work was published in Chemical Engineering Journal (IF: 13.3) and ACS Sustainable Chemistry & Engineering (IF: 8.4). His expertise extends to Graph Neural Networks for ADMET prediction and rational design of lipid-prodrug bioconjugates.

Dr. Mottafegh obtained his B.Sc. in Chemical Engineering from Sharif University of Technology (Iran, 2017), the country's premier engineering institution, ranking 106th among ~55,000 engineering candidates (top 0.2%) in the National University Entrance Exam. He ranked in the top 10% of his cohort and completed his thesis on polymer composites for friction materials.

He served as Project Leader for a \$50,000 KICET grant (2022-2025), managing a team of 4 Master's students in designing SiC-based 3D-printed ceramic heat exchangers, and as Key Researcher for a Samsung Advanced Institute of Technology project (2021-2023) on flash precipitation automation. He established an international collaboration with CSIR-Indian Institute of Chemical Technology resulting in 4 co-authored publications. He holds 3 patent applications covering scalable mRNA production platforms and AI-guided chromatography systems.

Dr. Mottafegh has published 21 peer-reviewed papers (12 as first/co-first author) and 3 in process patents, garnered > 167 citations (h-index 7, i10-index 3), and received outstanding publication awards at POSTECH for two consecutive years (2023-2024), the Best Poster Award at the 2023 KICHE Conference, and served as Invited Speaker at CKC 2025 (Montreal) and KFLEX 2025 (Daejeon), and Academic Mentor for the 2025 KRICT ChemDX Hackathon.

Part C. RELEVANT MERITS

C.1. Fellowships and Awards

- [1]. Juan de la Cierva Fellowship (2025) — Awarded by the Spanish Ministry of Science, hosted at IMDEA Materials Institute, Spain
- [2]. MSCA COFUND Postdoctoral Fellowship, BS4S (2026) — Awarded by European Union, hosted at Institute of Physical Chemistry PAS (ICChF), Poland
- [3]. Western University Postdoctoral Fellowship (Internal, 2026) — Awarded, Western University at Ontario, Canada

C.2. Publications

Under Review

- [1]. G. Ahn[†], **A. Mottafegh**, DALI: Diffusion-based Active Learning for Inverse Design regarding Flexible Optimization of Chemical Processes. 2026, **Under Review**

First & Co-First Authored

- [2]. **A. Mottafegh** and G. Ahn*. 2026, FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design, **Preprint**. (<https://doi.org/10.26434/chemrxiv.15003445/v1>).
- [3]. **A. Mottafegh*** and G. Ahn*. 2026, Adaptive Human-in-the-Loop Optimization Using Language-Guided Priors for Chemical Synthesis, **Journal of Computational Information and Modelling** (Accepted).
- [4]. V. Sharma[†], **A. Mottafegh**[†], J.U. Joo[†], D.P. Kim*. 2026, Integrated Microfluidic Platform for Quasi-Continuous High-Efficiency mRNA Production, **Lab on a Chip**, 26, 2923-2936. (<https://doi.org/10.1039/d6lc00118a>). Development of a fully automated flow platform for end-to-end manufacturing of the mRNA biomolecules.
- [5]. R. Chauhan[†], **A. Mottafegh**[†], A. Rana[†], D.P. Kim*, A.K. Singh*. 2026, UV-Visible Auto-Reactometry Enabled Parallel Screening for Accelerating Chemical Synthesis Development with LLM, **Chemical Engineering Journal**, 174829. (DOI:10.1016/j.cej.2026.174829). Utilization of the LLMs combined with BO in an SDL for parallelized flow synthesis.
- [6]. **A. Mottafegh**, A. Rohimmahtunissa, D.P. Kim*. 2025, Rapid prototyping of 3D-printed microchannel heat exchangers via AI-driven design and optimization. **Chemical Engineering Journal**, 503:166129. (DOI: 10.1016/J.CEJ.2025.166129, cited by 3). Application of generative design and AI algorithms to optimize the geometry and thermal efficiency of 3D-printed micro-heat exchangers.
- [7]. **A. Mottafegh**[†], J.U. Joo[†], G.S. Na[†], V. Sharma, D.P. Kim*. 2024, AI-assisted autonomous manufacturing of tailored drug-loaded nanoparticles by multi-step continuous-flow platform. **Chemical Engineering Journal**, 500:157454. (DOI: 10.1016/J.CEJ.2024.157454, cited by 16). Development of a fully autonomous platform integrating microfluidics and Bayesian optimization for tailored nanoparticle synthesis, reducing optimization time to ~20 mins while achieving >90% encapsulation efficiency.

- [8]. **A. Mottafegh**, J. Yang, G.S. Na, D.P. Kim*. 2024, Self-Driving Synthesis of Protein Nanoparticles by Active Transfer-Learning-Assisted Autonomous Flow Platform. **ACS Sustainable Chemistry & Engineering**, 12(35):13185-13196. (DOI: [10.1021/ACSSUSCHEMENG.4C03592](https://doi.org/10.1021/ACSSUSCHEMENG.4C03592), cited by 2). Implementation of Transfer Learning strategies to accelerate the discovery of protein nanoparticles with minimal experimental datasets.
- [9]. V. Sharma†, **A. Mottafegh**†, J.U. Joo, J.H. Kang, L. Wang, D.P. Kim*. 2024, Toward microfluidic continuous-flow and intelligent downstream processing of biopharmaceuticals. **Lab on a Chip**, 24(11):2861-2882. (DOI: [10.1039/D3LC01097J](https://doi.org/10.1039/D3LC01097J), cited by 45). Establishment of an intelligent continuous-flow platform for the downstream processing and purification of biopharmaceuticals.
- [10]. V. Sharma†, J.U. Joo†, **A. Mottafegh**†, D.P. Kim*. 2024, Continuous and autonomous-flow separation of laccase enzyme utilizing functionalized aqueous two-phase system with computer vision control. **Biore-source Technology**, 403:130888. (DOI: [10.1016/J.BIO_RTECH.2024.130888](https://doi.org/10.1016/J.BIO_RTECH.2024.130888), cited by 7). Integration of computer vision systems to automate the separation of enzymes in aqueous two-phase systems.
- [11]. S. Lee†, **A. Mottafegh**†, D.P. Kim*. 2024, Parameter investigation of an organic-inorganic hybrid resin for a 3D-printed microchannel heat exchanger. **Reaction Chemistry & Engineering**, 9:2089-2097. (DOI: [10.1039/D3RE00694H](https://doi.org/10.1039/D3RE00694H), cited by 2). Characterization of novel hybrid resins to enhance the thermal and mechanical stability of 3D-printed microreactors.
- [12]. **A. Mottafegh**†, G.N. Ahn†, D.P. Kim*. 2023, Meta optimization based on real-time Benchmarking of multiple surrogate models for autonomous flow synthesis. **Lab on a Chip**, 23(6):1613-1621. (DOI: [10.1039/D2LC00938B](https://doi.org/10.1039/D2LC00938B), cited by 11). Meta-learning study of multiple surrogate models to enhance the efficiency of autonomous algorithms in flow synthesis.

Co-Authored

- [13]. V. Sharma, J.U. Joo, **A. Mottafegh**, D.P. Kim*. 2025, Oscillatory Microfluidic Bioreactor for Accelerated mRNA Synthesis and Enhanced Protein Expression. **Chemical Engineering Journal**, 501:164284. (DOI: [10.1016/J.CEJ.2025.164284](https://doi.org/10.1016/J.CEJ.2025.164284), cited by 2).
- [14]. S.H. Pyen, **A. Mottafegh**, D.P. Kim*. 2025, Microfluidic Generation of Drug-loaded Polymer Microcapsules with Modulated Core-Shell Morphologies and Drug Release Profiles. **Chemical Engineering Journal**, 515:163244. (DOI: [10.1016/J.CEJ.2025.163244](https://doi.org/10.1016/J.CEJ.2025.163244), cited by 3).
- [15]. R. Chauhan, A. Rana, **A. Mottafegh**, D.P. Kim*, A.K. Singh*. 2025, Manual to Auto-Optimization Platform of Multi-step Apixaban Synthesis. **Organic Process Research & Development**, 29(3):881-888. (DOI: [10.1021/ACS.OPRD.4C00535](https://doi.org/10.1021/ACS.OPRD.4C00535), cited by 5).
- [16]. J.U. Joo, G.S. Na, V. Sharma, **A. Mottafegh**, D.P. Kim*. 2024, Ultrafast continuous flow-dialysis for nanoparticle-based drug delivery systems via microfluidic-multiple buffer injector. **Chemical Engineering Journal**, 500:156631. (DOI: [10.1016/J.CEJ.2024.156631](https://doi.org/10.1016/J.CEJ.2024.156631), cited by 7).
- [17]. A. Rana, R. Chauhan, **A. Mottafegh**, D.P. Kim*, A.K. Singh*. 2024, DigiChemTree enables programmable light-induced carbene generation for on demand chemical synthesis. **Communications Chemistry**, 7(1):251. (DOI: [10.1038/S42004-024-01330-Z](https://doi.org/10.1038/S42004-024-01330-Z), cited by 4).
- [18]. D. Aand, A. Rana, **A. Mottafegh**, D.P. Kim*, A.K. Singh*. 2024, Autonomous closed-loop photochemical reaction optimization for the synthesis of various angiotensin II receptor blocker molecules. **Reaction Chemistry & Engineering**, 9(9):2427-2435. (DOI: [10.1039/D4RE00138A](https://doi.org/10.1039/D4RE00138A), cited by 6).
- [19]. M. Purwa, G. Chandrakanth, A. Rana, **A. Mottafegh**, S. Kumar, D.P. Kim*, A.K. Singh*. 2024, Auto-Optimized Electro-Flow Reactor Platform for the in-situ Reduction of P(V) Oxide to P(III) and Their Application. **Chemistry–An Asian Journal**, 19(18):e202400438. (DOI: [10.1002/ASIA.202400438](https://doi.org/10.1002/ASIA.202400438), cited by 2).
- [20]. S.A. Kawale, D.C. Kang, G.N. Ahn, **A. Mottafegh**, J.H. Kang, G.S. Na, D.P. Kim*. 2023, Rapid flow synthesis of fenofibrate via scalable flash chemistry with in-line Li recovery. **Chemical Engineering Journal**, 477:147033. (DOI: [10.1016/J.CEJ.2023.147033](https://doi.org/10.1016/J.CEJ.2023.147033), cited by 4).
- [21]. S.J. Yim, K.W. Gyak, S.A. Kawale, **A. Mottafegh**, C.H. Park, Y. Ko, I. Kim, S.S. Jee, D.P. Kim*. 2023, One-flow multi-step synthesis of a monomer as a precursor of thermal-conductive semiconductor packaging polymer via multi-phasic separation. **Journal of Industrial and Engineering Chemistry**, 123:41-50. (DOI: [10.1016/J.JIEC.2023.03.018](https://doi.org/10.1016/J.JIEC.2023.03.018), cited by 1).
- [22]. M. Outokesh, **A. Mottafegh**, A. Nouri-Borujerdi*, S. Dolati. 2023, An Experimental-Theoretical study on heat transfer aspects of indirect Sublimation, for application in CO₂-based refrigeration systems. **Thermal Science and Engineering Progress**, 42:101881. (DOI: [10.1016/J.TSEP.2023.101881](https://doi.org/10.1016/J.TSEP.2023.101881), cited by 2)

C.3. Patents

- [1]. D.P. Kim, V. Sharma, **A. Mottafegh**. AI-Guided Autonomous Chromatography Platform for On-Demand Buffer Generation and Real-Time Process Monitoring in Biomolecule Purification. PCT International Application, IP25-00280 (2025.07.01), POSTECH.
- [2]. D.P. Kim, V. Sharma, J.U. Joo, **A. Mottafegh**. Scalable Miniaturized Integrated Platform for mRNA Therapeutics Production. PCT International Application, 10-2024-0194382 (2024.12.23), POSTECH.
- [3]. D.P. Kim, **A. Mottafegh**. Continuous Flow-Based Integrated Artificial Intelligence Microfluidic Platform for Optimized Drug Delivery Nanoparticle Production. US Patent Application, 10-2024-0138638 (2024.06.07), POSTECH.

C.4. Conference Presentations

Oral Presentations

- [1]. AI4X Acceleration Conference - **Oral Presentation**, Singapore, 2026
- [2]. Canada Korea Conference in Science and Technology (CKC 2025) - **Invited Speaker**, Montreal, Canada, 2025
- [3]. Korean Flow Chemistry Exchange Meeting (KFLEX 2025) - **Invited Speaker**, Daejeon, South Korea, 2025
- [4]. Korean International Chemical Engineering Conference (KICHE) - **Oral Presentation**, Busan, South Korea, 2023,

Poster Presentations

- [5]. World Biomaterial Congress (WBC 2024) - **Poster Presentation**, Daegu, South Korea, 2024
- [6]. Materials Research Society of Korea (MRS) - **Poster Presentation**, Jeju, South Korea, 2024
- [7]. Korean Society of Pharmaceutical Technology (KSPST) - **Poster Presentation**, Seoul, South Korea, 2024
- [8]. International Conference on High Temperature Ceramic Matrix Composites (HT-CMC24) - **Poster Presentation**, Jeju, South Korea, 2023

Others

- [9]. KRICT ChemDX Hackathon 2025 - **Academic Mentor**, Seoul, South Korea, 2025

C.5. Research Projects

- [1]. 2023.06 – 2025.05 Design and Formulation of SiC-Based 3D-Printed Ceramic Heat Exchangers, Korean Institute of Ceramic Engineering and Technology (KICET), **South Korea**
Project Leader (\$50,000 USD). The applicant served as the technical lead for this project, co-authoring the grant proposal and supervising a team of 4 Master's students. He was responsible for developing high-performance SiC ceramic heat exchangers using stereolithography 3D printing techniques. This work successfully delivered a prototype exhibiting superior thermal conductivity and mechanical stability for industrial applications (resulting in publication C.1 [4], [9]).
- [2]. 2021.10 – 2023.05 Flash Precipitation of Random Copolymers for Generation of Uniform Nanoparticles Using a 3D Flow-Focusing Micro-Mixer, Samsung Advanced Institute of Technology (SAIT), **South Korea**
Key Researcher (\$120,000 USD). In this industrial collaboration with Samsung, the applicant was responsible for engineering the automation architecture for the flash precipitation process. He designed the 3D flow-focusing microfluidic setup and implemented real-time control algorithms to ensure the uniformity of random copolymer nanoparticles. The optimized protocols developed in this project were transferred to the industrial partner for scaling up material production.